

VIET NAM NATIONAL UNIVERSITY HO CHI MINH CITY

UNIVERSITY OF SCIENCE

FACULTY OF INFORMATION TECHNOLOGY



Report

Bonus Assignment

Course: Introduction to Software Engineering

Supervisors:

Nguyễn Văn Vũ

Student:

23127230 - Đặng Lữ Kiều My

23127266 - Nguyễn Anh Thư

January 10, 2026

1 Setup and Evaluation Agent Configuration

This section provides a comprehensive guide for setting up the evaluation environment for the web-based agent system using SeeAct, Ollama, and Docker.

Download source code setup from: https://drive.google.com/drive/folders/1LCCPwA54lo4LjoZfBTP_Ds_3Ad17W5IP?usp=drive_link

1.1 System Requirements

- **Operating System:** Windows 10/11, macOS
- **Python:** Version ≥ 3.11
- **Docker Desktop:** Installed and running
- **Git:** For cloning the repository (if not already installed)

1.2 Python Dependencies Installation

1.2.1 Create Virtual Environment (Recommended)

```
1 # Create virtual environment
2 python -m venv venv
3
4 conda activate venv
```

1.2.2 Install Required Packages

Install the main dependencies:

```
1 pip install playwright
```

Install the SeeAct package from source (if needed):

```
1 cd AGENT\SeeAct\seeact_package
2 pip install -e .
3 cd ..\..\..\..
```

Install Playwright browsers:

```
1 playwright install chromium
```

Note: If encountering errors installing **torch** on Windows, use:

```
1 pip install torch --index-url
2 https://download.pytorch.org/whl/cpu
```

1.3 Setting Up Ollama and LLaVA Model

1.3.1 Install Ollama

1. Download Ollama from: <https://ollama.ai/download>
2. Install and run Ollama Desktop (or service)

1.3.2 Pull LLaVA Model

Pull the recommended LLaVA model:

```
1 # Pull model llava (recommended: llava:7b or llava:13b
2 # depending on available RAM)
3 ollama pull llava
4
5 # Verify model installation
6 ollama list
```

1.4 Docker Environment Setup

1.4.1 Ensure Docker Desktop is Running

Open **Docker Desktop** and verify its status (the Docker icon in the system tray should be green).

1.4.2 Configure Docker for PrestaShop

PrestaShop is configured in `environments/PrestaShop/docker-compose.yml`.

Default Configuration:

- **Port:** 8080:80 (accessible at `localhost:8080`)
- **Database:** MariaDB 10.5
- **Auto-install:** Enabled
- **Admin email:** `admin@prestashop.com`
- **Admin password:** `password123456789`

1.4.3 Test Docker Compose (Optional)

```
1 cd environments\PrestaShop
2 docker compose up -d
3
4 # Check containers
```

```
5 docker ps
6
7 # Check logs (wait ~2-3 minutes for PrestaShop
8 # installation to complete)
9 docker compose logs -f prestashop
10
11 # Test URL: Open browser at http://localhost:8080
12
13 # Stop containers (if needed)
14 docker compose down
15 cd ..\..
```

Note: This step is optional if `run_webeval.py` automatically starts the containers.

1.5 Task and Environment Configuration

1.5.1 Task Configuration File: `dataset/tasks.json`

This file contains the list of tasks to evaluate. Each task follows this format:

```
1 {
2   "task_id": "ps_01",
3   "task_description": "Subscribe to the website via
4     the footer subscription field with email
5     'account.test.4@mynes.com'.",
6   "task_type": "operation",
7   "start_url": "__PRESTASHOP__",
8   "require_login": false,
9   "eval": {
10     "eval_type": ["dom_match"],
11     "dom_match": {
12       "url": "last",
13       "dom_extractor": "document.querySelector(
14         '.alert.alert-success').innerText",
15       "match_type": "contains",
16       "match_value": "You have successfully subscribed
17         to this newsletter.",
18       "description": "Verify that the success message
19         is displayed after subscribing."
20     }
21   }
22 }
```

Important Notes:

- `start_url` must match a key in `environments/env_config.json` (e.g., `__PRESTASHOP__`)
- `eval_type` can be: `dom_match`, `url_match`, `string_match`, or `semantic_match`
- The `docker_env` field has been removed (the script automatically infers from `start_url`)

1.5.2 Environment Configuration File: `environments/env_config.json`

This file maps `start_url` keys to actual URLs and credentials:

```
1 {
2   "__PRESTASHOP__": {
3     "url": "http://localhost:8080",
4     "username": "admin@prestashop.com",
5     "password": "password123456789"
6   }
7 }
```

1.6 SeeAct Runner Configuration

In `agents/seeact_runner.py`, the default configuration is:

```
1 SeeActRunnerConfig(
2     model="llava",    # Use local Ollama LLaVA
3     max_op=10,        # Maximum number of actions
4     monitor=False,    # Do not pause before each action
5     top_k=3,          # Number of choices for LLM to select
6 )
```

1.7 Running the Benchmark

1.7.1 Run All Tasks

Ensure you are in the root directory:

```
1 # Run all tasks
2 python run_webeval.py
3
4 # Or specify a specific tasks file
5 python run_webeval.py --tasks dataset/tasks.json
6
7 # Or specify environment config
```

```
8 python run_webeval.py
9     --env-config environments/env_config.json
```

1.8 Results and Logs

1.8.1 Console Output

The script will output:

- Task ID and description
- Docker compose status
- SeeAct agent actions
- Evaluation results (PASS/FAIL)
- Final TSR (Task Success Rate): `passed/total`

1.8.2 SeeAct Output Files

SeeAct saves screenshots and logs in:

```
1 AGENT/SeeAct/seeact_agent_files/{timestamp}/
2 |- screenshots/
3 |   |- screen_1.png
4 |   |- ...
5 |- terminal_output.txt
```

1.9 Installation Resources

The following tools and resources are required:

- **SeeAct:** <https://github.com/OSU-NLP-Group/SeeAct>
- **Ollama:** <https://ollama.ai>
- **Playwright:** <https://playwright.dev/python>
- **Docker Compose:** <https://docs.docker.com/compose>

Table 1: PRESTASHOP Site Test Tasks

task_id	task_description	task_type	start_url	require_login	eval	result
ps_01	Subscribe to the website via the footer subscription field with email 'account.test.4@mynes.com'.	operation	__PRESTASHOP --	false	type: dom_match extractor: document.querySelector('.alert.alert-success').innerText match: contains "You have successfully subscribed to this newsletter." desc: Verify that the success message is displayed after subscribing.	Pass
ps_02	Search for product 'Printed Dress' and verify result contains the product name.	operation	__PRESTASHOP --	false	type: dom_match extractor: document.querySelector('.product-title a').innerText match: contains "The Best Is Yet To Come" desc: Search results should include 'The Best Is Yet To Come'.	Pass
ps_03	Open a product page for 'Printed Dress' and add it to cart; verify cart success message.	operation	__PRESTASHOP --	false	type: dom_match extractor: document.querySelector('#blockcart-modal .modal-title').innerText match: contains "Product successfully added to your shopping cart" desc: Confirm product added to cart success message.	Pass
ps_04	Add a product to the cart from the category listing and verify cart count increased.	operation	__PRESTASHOP --	false	type: dom_match extractor: document.querySelector('.cart-products-count').innerText match: contains "(1)" desc: Verify the cart quantity displays at least '1' after adding.	Pass
ps_05	Proceed to checkout as guest and verify the checkout page heading is shown.	operation	__PRESTASHOP --	false	type: dom_match extractor: document.querySelector('h1').innerText match: contains "PERSONAL INFORMATION" desc: Verify the checkout page is reached (Section 1 Title).	Fail
ps_06	Create a new account with email 'new.user.test1@mynes.com' and verify account creation success message.	operation	__PRESTASHOP --	false	type: dom_match extractor: document.querySelector('.account-created, .alert-success, .confirmation')?.innerText " match: contains "Your account has been created" desc: Confirm account creation success text is displayed.	Fail

Continued on next page...

Table 1 – continued from previous page

task_id	task_description	task_type	start_url	require_login	eval	result
ps_07	Log in with an existing account and verify user name appears in header.	operation	__PRESTASHOP --	false	type: dom_match extractor: document.querySelector('.account, .user-info .username, .header_user_info .user-name')?.innerText " match: contains "New User" desc: Verify logged-in username or welcome message appears in header.	Pass
ps_08	Use the contact form to send a message and verify confirmation is displayed.	operation	__PRESTASHOP --	false	type: dom_match extractor: document.querySelector('.contact_confirmation, .alert.alert-success, .contact-form .confirmation')?.innerText " match: contains "Your message has been successfully sent" desc: Contact form success confirmation should appear.	Pass
ps_09	Change the site currency to 'Euro' and verify the currency symbol updates on product prices.	operation	__PRESTASHOP --	false	type: dom_match extractor: document.querySelector('.product-price, .price')?.innerText " match: contains "€" desc: Product price should display the Euro symbol after switching currency.	Pass
ps_10	Filter a category by 'Size: M' and verify at least one product shows the chosen size availability.	operation	__PRESTASHOP --	false	type: dom_match extractor: Array.from(document.querySelectorAll('.product-variants, .product-sizes, .variant-list')).map(e=>e.innerText).join(' ') match: contains "M" desc: Products listed should show size 'M' available.	Pass
ps_11	Sort a product listing by 'Price: Lowest' and verify results contain price values (presence of price text).	operation	__PRESTASHOP --	false	type: dom_match extractor: document.querySelector('.product-price, .price')?.innerText " match: contains "\$" desc: Product prices should be visible after sorting (presence of price text).	Pass
ps_12	Add product to wishlist (requires login) and verify wishlist contains the product.	operation	__PRESTASHOP --	true	type: dom_match extractor: document.querySelector('.wishlist .product-name, .favorite-list .product-name')?.innerText " match: contains "Printed Dress" desc: Wishlist should list the chosen product after adding.	Fail

Continued on next page...

Table 1 – continued from previous page

task_id	task_description	task_type	start_url	require_login	eval	result
ps_13	Compare two products and verify the comparison table shows both product names.	operation	__PRESTASHOP --	false	type: dom_match extractor: Array.from(document.querySelectorAll('.compare-list .product-name, .comparison-table .product-name')).map(e=>e.innerText).join(' ') match: contains "Printed Dress" desc: Comparison page should display product names (at least one expected).	Pass
ps_14	Submit a product review (requires login) and verify the review confirmation message.	operation	__PRESTASHOP --	true	type: dom_match extractor: document.querySelector('.review-success, .comment-result, .alert-success')?.innerText " match: contains "Your review has been added" desc: Verify product review submission confirmation.	Fail
ps_15	Apply a coupon code during checkout and verify the order summary shows the discount.	operation	__PRESTASHOP --	false	type: dom_match extractor: document.querySelector('.cart-discounts, .discount-applied, .cart-summary .discount')?.innerText " match: contains "Discount" desc: Order summary should reflect the applied coupon/discount.	Fail
ps_16	Navigate to a category page and verify the category title is displayed.	operation	__PRESTASHOP --	false	type: dom_match extractor: document.querySelector('.category-name, .page-heading, h1')?.innerText " match: contains "CLOTHES" desc: Category page heading or name should be visible.	Pass
ps_17	Switch site language to French and verify a known French string appears (e.g., 'Mon compte').	operation	__PRESTASHOP --	false	type: dom_match extractor: document.querySelector('body')?.innerText " match: contains "Mon compte" desc: Switching language should show French text like 'Mon compte'.	Pass
ps_18	Use the 'Contact us' menu to open customer service and verify the support page title.	operation	__PRESTASHOP --	false	type: dom_match extractor: document.querySelector('#main h1, .contact-title, .page-heading')?.innerText " match: contains "Customer service" desc: Customer service/contact page heading should be present.	Pass

Continued on next page...

Table 1 – continued from previous page

task_id	task_description	task_type	start_url	require_login	eval	result
ps_19	View product images gallery and verify the gallery shows at least one larger image element.	operation	__PRESTASHOP --	false	type: dom_match extractor: document.querySelector('.product-main .thumb, .product-cover img, .product-image-photo')?.getAttribute('src') " match: contains ".jpg" desc: Product gallery should contain an image src (e.g., .jpg).	Pass
ps_20	Sign out from an account and verify the login link/button is shown again.	operation	__PRESTASHOP --	true	type: dom_match extractor: document.querySelector('#__desktop_user_info .user-info a, .login, .login-link')?.innerText " match: contains "Sign in" desc: After logout, the sign-in/login link or label should be visible.	Pass

Evaluation Summary:

The evaluation was conducted on 20 PrestaShop tasks using the WebAppEval benchmark framework. The evaluation employs `dom_match` matchers to automatically verify task completion by extracting DOM elements using JavaScript selectors and checking for expected text content using the `contains` match type.

Results:

- **Total Tasks:** 20
- **Passed:** 15 tasks (75.0%)
- **Failed:** 5 tasks (25.0%)
- **Task Success Rate (TSR):** 0.75

View on Agent Capabilities for End-to-End Testing:

Based on the evaluation results, while the agent demonstrates promising capabilities for automating simple to moderate complexity test scenarios (achieving 75% success rate), it currently falls short of fully replacing human testers for end-to-end testing, particularly for complex multi-step workflows that require understanding of intricate business logic and form validation. The agent excels at straightforward navigation, search, and basic interactions but struggles with tasks involving multi-field form submissions, checkout processes, and authentication-dependent complex workflows. Therefore, the agent can serve as a valuable complement to manual testing efforts, handling repetitive and routine test cases efficiently, while human testers remain essential for comprehensive test coverage, edge case validation, and scenarios requiring contextual understanding and decision-making.